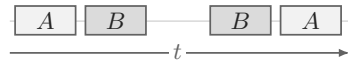


One recording x

$A = \text{dog}, B = \text{rooster}$



$W_+ : A \rightarrow B$

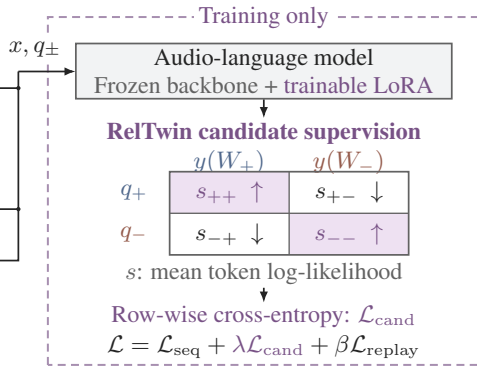
$W_- : B \rightarrow A$

q_+ : dog followed by rooster

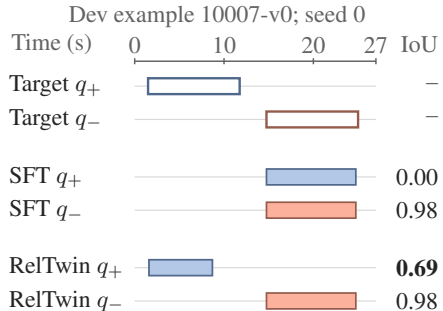
q_- : rooster followed by dog

Both relations hold in x ;
each query has a different target.

(a) Inverse-relation pairs



(b) Paired answer supervision



(c) Query-specific localization